Table S1. Study selection criteria

PECO elements	PECO question formulation*: Among humans diagnosed with a cancer predisposing syndrome (CPS), immunodeficiency, genetic mutation what is the impact on cancer risk estimates following low-to-moderate doses of ionizing radiation (IR) during childhood?
Population (P)	Studies involving humans with a cancer predisposing syndrome (CPS) immunodeficiency, genetic mutation exposed to ionizing radiation during childhood or young/early adulthood (<25 years).
Exposure (E)	Studies evaluating low (<100 mGy) to moderate (<500 mGy) IR exposure IR in the form of: medical, accidental, atomic bomb and/or environmental
Comparator (C)	N/A
Outcome (O)	Haematological malignancy (leukaemia and lymphoma), breast, thyroid, and brain tumors Cancer risk estimates: Relative risks (RR), Standardized incidence ratio (SIR) by level of dose, Excess relative risk / dose level Studies looking at second malignancies after primary cancer treatment Studies assessing radiotherapy should assess secondary cancers and not 3rd or 4th.
Study design	Cohort or case control Letters responding to included studies Pilot studies

* Formulation of the PECO question follows recent recommendations by Morgan et al., 2018

Table S2. Search strings used to perform literature search.

Search string	Database	Results	Date
(TITLE-ABS-KEY (predisposing) OR TITLE-ABS-KEY ("Radiation sensitivity") OR TITLE-ABS-KEY ("Radiosensitivity") OR TITLE-ABS-KEY ("Genetic disorder") OR TITLE-ABS-KEY ("Genetic disease") OR TITLE-ABS-KEY ("Genetic susceptibility") OR TITLE-ABS-KEY ("tumor suppressor gene") OR TITLE-ABS-KEY ("BRCA1") OR TITLE-ABS-KEY ("BRCA2") OR TITLE-ABS-KEY ("Pathophysiologic predisposition") OR TITLE-ABS-KEY ("Predisposing Cancer Syndrome") OR TITLE-ABS-KEY ("Hereditary Cancer Syndrome") OR TITLE-ABS-KEY ("Hereditary Neoplastic Syndrome") OR TITLE-ABS-KEY ("Birth Defect") OR TITLE-ABS-KEY ("Congenital Abnormality") OR TITLE-ABS-KEY (down* AND syndrome) OR TITLE-ABS-KEY ("Mongolism") OR TITLE-ABS-KEY ("47, XY, +21") OR TITLE-ABS-KEY ("Trisomy G") OR TITLE-ABS-KEY ("47, XX, +21") OR TITLE-ABS-KEY ("Trisomy 21") OR TITLE-ABS-KEY ("Noonan Syndrome") OR TITLE-ABS-KEY ("Turner syndrome") OR TITLE-ABS-KEY (" Kostmann granulocytopenia ") OR TITLE-ABS-KEY (" Nevoid basal cell syndrome ") OR TITLE-ABS-KEY ("Wiskott Aldrich Syndrome") OR TITLE-ABS-KEY (imd2) OR TITLE-ABS-KEY ("Other phacomatoses") OR TITLE-ABS-KEY ("Large granular lymphocytes ") OR TITLE-ABS-KEY ("Immunodeficiency 2") OR TITLE-ABS-KEY ("Ataxia Telangiectasia") OR TITLE-ABS-KEY ("ATM Gene") OR TITLE-ABS-KEY ("Louis Bar Syndrome") OR TITLE-ABS-KEY ("Shwachman Syndrome") OR TITLE-ABS-KEY ("Bloom Syndrome") OR TITLE-ABS-KEY (monosomy 7 AND bone AND marrow) OR TITLE-ABS-KEY ("Myelodysplasia monosomy 7") OR TITLE-ABS-KEY ("Chromosomes, Human, Pair 7") OR TITLE-ABS-KEY ("Myelodysplastic syndrome") OR TITLE-ABS-KEY ("Chromosome Disorder") OR TITLE-ABS-KEY (fanconi* AND anemia) OR TITLE-ABS-KEY ("Fanconi Hypoplastic Anemia") OR TITLE-ABS-KEY ("Bone Marrow Transplantation") OR TITLE-ABS-KEY ("Haematopoietic Stem Cell") OR TITLE-ABS-KEY ("Antibody Deficiency Syndrome") OR TITLE-ABS-KEY (li AND fraumeni* AND syndrome) OR TITLE-ABS-KEY ("Nijmegen breakage syndrome") OR TITLE-ABS-KEY (nbs)) OR (TITLE-ABS-KEY (neurofibromatoses) OR TITLE-ABS-KEY ("Neurofibromatoses type 1") OR TITLE-ABS-KEY (nf1) OR TITLE-ABS-KEY ("Molluscum Fibrosum") OR TITLE-ABS-KEY ("Neurofibromatoses type 2") OR TITLE-ABS-KEY (nf2) OR TITLE-ABS-KEY ("Tuberous Sclerosis") OR TITLE-ABS-KEY ("Cerebral Sclerosis") OR TITLE-ABS-KEY ("Tuberous Sclerosis") OR TITLE-ABS-KEY (klinefelter* AND syndrome) OR TITLE-ABS-KEY ("XXY Syndrome") OR TITLE-ABS-KEY ("48, XXYY Syndrome") OR TITLE-ABS-KEY ("49, XXXXY Syndrome") OR TITLE-ABS-KEY ("XXXY Male") OR TITLE-ABS-KEY ("Basal Cell Nevus Syndrome") OR TITLE-ABS-KEY ("Gorlin Syndrome") OR TITLE-ABS-KEY ("von Hippel-Lindau Disease") OR TITLE-ABS-KEY ("VHL Syndrome") OR TITLE-ABS-KEY ("Turcot Syndrome") OR TITLE-ABS-KEY ("MMR Deficiency") OR TITLE-ABS-KEY ("Mismatch Repair Deficiency") OR TITLE-ABS-KEY ("Mismatch Repair Cancer Syndrome") OR TITLE-ABS-KEY ("HIV") OR TITLE-ABS-KEY ("Human Immunodeficiency Virus") OR TITLE-ABS-KEY ("AIDS Virus") OR TITLE-ABS-KEY ("Acquired Immune Deficiency Syndrome Virus") OR TITLE-ABS-KEY ("Severe combined immune deficiency") OR TITLE-ABS-KEY ("Severe Combined Immunodeficiency") OR TITLE-ABS-KEY ("Organ Transplantation") OR TITLE-ABS-KEY ("Transplant Donor") OR TITLE-ABS-KEY ("Tissue Transplantation") OR TITLE-ABS-KEY ("Transplant Recipient") OR TITLE-ABS-KEY ("Common variable immune deficiency") OR TITLE-ABS-KEY ("Congenital cardiovascular anomaly") OR TITLE-ABS-KEY (	SCOPUS	630 1005 1715 1508	2019-11-06 2020-01-28 2020-02-12 2020-02-12

"Congenital heart defects") OR TITLE-ABS-KEY ("Nervous System Malformation") OR TITLE-ABS-KEY ("Central nervous system malformation") OR TITLE-ABS-KEY ("Nervous System Abnormality") OR TITLE-ABS-KEY ("Congenital Nervous System Malformation") OR TITLE-ABS-KEY ("Birth defect") OR TITLE-ABS-KEY ("Congenital Abnormality") OR TITLE-ABS-KEY ("Congenital Anamoly") OR TITLE-ABS-KEY ("Congenital Malformation")) OR (TITLE-ABS-KEY ("Congenital Defect") OR TITLE-ABS-KEY ("22q11.2 deletion syndrome") OR TITLE-ABS-KEY ("DiGeorge Syndrome") OR TITLE-ABS-KEY ("Third and Fourth Pharyngeal Pouch Syndrome") OR TITLE-ABS-KEY ("Velocardiofacial Syndrome") OR TITLE-ABS-KEY ("Sedlackova Syndrome") OR TITLE-ABS-KEY ("Shprintzen Syndrome") OR TITLE-ABS-KEY (rasopathies) OR TITLE-ABS-KEY ("Langerhans cell histiocytosis") OR TITLE-ABS-KEY ("Immunodeficiency syndrome")) AND (TITLE-ABS-KEY (radiation) TITLE-ABS-KEY ("Ionizing Radiation") OR TITLE-ABS-KEY ("Radiation accident") OR TITLE-ABS-KEY ("Nuclear Accident") OR TITLE-ABS-KEY ("Nuclear Reactor Accident") OR TITLE-ABS-KEY ("Background radiation") OR TITLE-ABS-KEY ("Natural Radiation") OR TITLE-ABS-KEY ("Medical radiation") OR TITLE-ABS-KEY ("Atomic bomb") OR TITLE-ABS-KEY ("CT scan") OR TITLE-ABS-KEY ("CT imaging") OR TITLE-ABS-KEY ("Computed tomography") OR TITLE-ABS-KEY ("PET-CT scan") OR TITLE-ABS-KEY ("Nuclear medicine") OR TITLE-ABS-KEY ("Diagnostic radiation") OR TITLE-ABS-KEY (radiotherapy) OR TITLE-ABS-KEY ("Radiation treatment") OR TITLE-ABS-KEY ("PET scan") OR TITLE-ABS-KEY ("Medical screening") OR TITLE-ABS-KEY ("Occupational radiation")) AND (TITLE-ABS-KEY (leukaemia) OR TITLE-ABS-KEY (lymphoma) OR TITLE-ABS-KEY ("Haematologic Malignancy") OR TITLE-ABS-KEY ("Haematologic Neoplasm") OR TITLE-ABS-KEY ("Brain tumour") OR TITLE-ABS-KEY ("Brain Cancer") OR TITLE-ABS-KEY ("Intracranial Neoplasm") OR TITLE-ABS-KEY ("Breast Tumour") OR TITLE-ABS-KEY ("Breast Cancer") OR TITLE-ABS-KEY ("Thyroid Tumour") OR TITLE-ABS-KEY ("Thyroid Cancer") OR TITLE-ABS-KEY ("Thyroid Carcinoma") OR TITLE-ABS-KEY ("Thyroid Adenoma")) AND (LIMIT-TO (EXACTKEYWORD , "Humans")) AND (LIMIT-TO (LANGUAGE , "English"))			
(rb1 AND mutation OR rb1 AND gene OR retinoblastoma AND gene AND radiotherapy AND retinoblastoma)	SCOPUS	93	2020-05-02
(((((((predisposing[Text Word] OR Radiation sensitivity[Text Word] OR Radiosensitivity[Text Word] OR Genetic disorder[Text Word] OR Genetic disease[Text Word] OR Genetic susceptibility[Text Word] OR tumor suppressor gene[Text Word] OR “BRCA1”[Text Word] OR “BRCA2”[Text Word] OR Pathophysiologic predisposition[Text Word] OR Predisposing Cancer Syndrome[Text Word] OR Hereditary Cancer Syndrome[Text Word] OR Hereditary Neoplastic Syndrome[Text Word] OR Birth Defect[Text Word] OR Congenital Abnormality[Text Word] OR Down* Syndrome[Text Word] OR Mongolism[Text Word] OR 47, XY, +21[Text Word] OR Trisomy G[Text Word])) OR (47, XX, +21[Text Word] OR Trisomy 21[Text Word] OR Noonan Syndrome[Text Word] OR Turner syndrome[Text Word] OR Wiskott Aldrich Syndrome[Text Word] OR Imd2[Text Word] OR Other phacomatoses[Text Word] OR granular lymphocytes[Text Word] OR Immunodeficiency 2[Text Word] OR Ataxia Telangiectasia[Text Word] OR ATM Gene[Text Word] OR Louis Bar Syndrome[Text Word] OR Shwachman Syndrome[Text Word] OR Bloom Syndrome[Text Word] OR Monosomy 7 AND Bone Marrow[Text Word] OR Myelodysplasia monosomy 7[Text Word] OR Chromosomes, Human, Pair 7[Text Word] OR Myelodysplastic syndrome[Text Word] OR Chromosome Disorder[Text Word] OR Fanconi* Anemia[Text Word] OR Fanconi Hypoplastic Anemia[Text Word] OR Bone Marrow Transplantation[Text Word] OR Haematopoietic Stem Cell[Text Word] OR Antibody Deficiency Syndrome[Text Word]	PubMed	2257 2800	2019-11-08 2020-02-07

OR Li Fraumeni* Syndrome[Text Word] OR Nijmegen breakage syndrome[Text Word])) OR (NBS [Text Word] OR Neurofibromatoses[Text Word] OR Neurofibromatoses type 1[Text Word] OR NF1[Text Word] OR Molluscum Fibrosum[Text Word] OR Neurofibromatoses type 2[Text Word] OR NF2[Text Word] OR Tuberous Sclerosis[Text Word] OR Cerebral Sclerosis[Text Word] OR Tuberous Sclerosis[Text Word] OR Klinefelter* Syndrome[Text Word] OR XXY Syndrome[Text Word] OR 48, XXYY Syndrome[Text Word] OR 49, XXXXY Syndrome[Text Word] OR XXXY Male[Text Word] OR Basal Cell Nevus Syndrome[Text Word] OR Gorlin Syndrome[Text Word] OR von Hippel-Lindau Disease[Text Word] OR VHL Syndrome[Text Word] OR Turcot Syndrome[Text Word] OR Kostmann granulocytopenia[Text Word] OR Nevoid basal cell syndrome[Text Word] OR MMR Deficiency[Text Word] OR Mismatch Repair Deficiency[Text Word] OR Mismatch Repair Cancer Syndrome[Text Word] OR HIV[Text Word] OR Human Immunodeficiency Virus[Text Word] OR AIDS Virus[Text Word])) OR (Acquired Immune Deficiency Syndrome Virus[Text Word] OR Severe combined immune deficiency[Text Word] OR Severe Combined Immunodeficiency[Text Word] OR Organ Transplantation[Text Word] OR Transplant Donor[Text Word] OR Tissue Transplantation[Text Word] OR Transplant Recipient[Text Word] OR Common variable immune deficiency[Text Word] OR Congenital cardiovascular anomaly[Text Word] OR Congenital heart defects[Text Word] OR Nervous System Malformation[Text Word] OR Central nervous system malformation[Text Word] OR Nervous System Abnormality[Text Word] OR Congenital Nervous System Malformation[Text Word] OR Birth defect[Text Word])) OR (Congenital Abnormality[Text Word] OR Congenital Anamoly[Text Word] OR Congenital Malformation[Text Word] OR Congenital Defect[Text Word] OR 22q11.2 deletion syndrome[Text Word] OR DiGeorge Syndrome[Text Word] OR “Third and Fourth Pharyngeal Pouch Syndrome"[Text Word] OR Velocardiofacial Syndrome[Text Word] OR Sedlackova Syndrome[Text Word] OR Shprintzen Syndrome[Text Word] OR rasopathies[Text Word] OR Langerhans cell histiocytosis[Text Word] OR Immunodeficiency syndrome[Text Word])) AND (radiation[Text Word] OR ionizing radiation[Text Word] OR radiation accident[Text Word] OR nuclear accident[Text Word] OR nuclear reactor accident[Text Word] OR background radiation[Text Word] OR natural radiation[Text Word] OR medical radiation[Text Word] OR atomic bomb[Text Word] OR x-ray computed tomography'[Text Word] OR ct scan[Text Word] OR ct imaging[Text Word] OR computed tomography[Text Word] OR pet-ct scanner[Text Word] OR pet-ct scan[Text Word] OR nuclear medicine[Text Word] OR diagnostic radiation[Text Word] OR radiotherapy[Text Word] OR radiation treatment[Text Word] OR pet scanner[Text Word] OR pet scan[Text Word] OR medical screening[Text Word] OR occupational radiation[Text Word]))) AND (leukaemia[Text Word] OR lymphoma[Text Word] OR haematologic malignancy[Text Word] OR haematologic disease[Text Word] OR haematologic neoplasm[Text Word] OR brain tumour[Text Word] OR breast cancer[Text Word] OR thyroid tumour[Text Word] OR thyroid cancer[Text Word] OR thyroid carcinoma[Text Word] OR thyroid adenoma[Text Word]) Sort by: Best Match Filters: Humans; English			
((((RB1 mutation[Title/Abstract]) OR (RB1 gene[Title/Abstract])) OR (Retinoblastoma gene[Title/Abstract])) AND (Radiotherapy[Title/Abstract])) AND (Retinoblastoma[Title/Abstract])	PubMed	37	2020-05-02
(TS=predisposing OR TS=”Radiation sensitivity” OR TS=Radiosensitivity OR TS=”Genetic disorder” OR TS=”Genetic disease” OR TS=”Genetic susceptibility” OR TS=”tumour suppressor gene" OR TS=BRCA1 OR TS=BRCA2 OR TS=”Pathophysiologic predisposition” OR TS=”Predisposing Cancer Syndrome” OR TS=”Hereditary Cancer Syndrome” OR TS=”Hereditary Neoplastic Syndrome” OR TS=”Birth Defect” OR TS=”Congenital Abnormality” OR TS=”Down* Syndrome” OR TS=”Mongolism” OR TS=”47, XY, +21” OR TS=”Trisomy G” OR	Web of Science	1459 577 500	2019-11-08 2020-01-28 2020-02-12

TS=”47, XX, +21” OR TS=”Trisomy 21” OR TS=”Noonan Syndrome” OR TS=”Turner syndrome” OR TS=”Wiskott Aldrich Syndrome” OR TS=Imd2 OR TS=”Other phacomatoses” OR TS=”granular lymphocytes” OR TS=”Immunodeficiency 2” OR TS=”Ataxia Telangiectasia” OR TS=”ATM Gene” OR TS=”Louis Bar Syndrome” OR TS=”Shwachman Syndrome”) OR (TS=”Bloom Syndrome” OR TS=("Monosomy 7" AND "Bone Marrow") OR TS=”Myelodysplasia monosomy 7” OR TS=("Chromosome" AND "Human" AND "Pair 7") OR TS=”Myelodysplastic syndrome” OR TS=”Chromosome Disorder” OR TS=”Fanconi* Anemia” OR TS=”Fanconi Hypoplastic Anemia” OR TS=”Bone Marrow Transplantation” OR TS=”Haematopoietic Stem Cell” OR TS=”Antibody Deficiency Syndrome” OR TS=”Li Fraumeni* Syndrome” OR TS=”Nijmegen breakage syndrome” OR TS=NBS OR TS=Neurofibromatoses OR TS=”Neurofibromatoses type 1” OR TS=NF1 OR TS=”Molluscum Fibrosum” OR TS=”Neurofibromatoses type 2” OR TS=NF2 OR TS=”Tuberous Sclerosis” OR TS=”Cerebral Sclerosis”) OR (TS="Klinefelter* Syndrome" OR TS=”XXY Syndrome” OR TS=”48, XXYY Syndrome” OR TS=”49, XXXXY Syndrome” OR TS=”XXXY Male” OR TS=”Basal Cell Nevus Syndrome” OR TS=”Gorlin Syndrome” OR TS=”von Hippel-Lindau Disease” OR TS=”VHL Syndrome” OR TS=”Turcot Syndrome” OR TS=”Kostmann granulocytopenia” OR TS=”Nevoid basal cell syndrome” OR TS=”MMR Deficiency” OR TS=”Mismatch Repair Deficiency” OR TS=”Mismatch Repair Cancer Syndrome” OR TS=HIV OR TS=”Human Immunodeficiency Virus” OR TS=”AIDS Virus” OR TS=”Acquired Immune Deficiency Syndrome Virus” OR TS=”Severe combined immune deficiency” OR TS=”Severe Combined Immunodeficiency” OR TS=”Organ Transplantation” OR TS=”Transplant Donor” OR TS=”Tissue Transplantation” OR TS=”Transplant Recipient” OR TS=”Common variable immune deficiency” OR TS=”Congenital cardiovascular anomaly” OR TS=”Congenital heart defects” OR TS=”Nervous System Malformation” OR TS=”Central nervous system malformation” OR TS=”Nervous System Abnormality” OR TS=”Congenital Nervous System Malformation” OR TS=”Birth defect” OR TS=”Congenital Abnormality" OR TS="Congenital Anamoly" OR TS="Congenital Malformation" OR TS="Congenital Defect” OR TS=”22q11.2 deletion syndrome” OR TS=”DiGeorge Syndrome” OR TS=“Third and Fourth Pharyngeal Pouch Syndrome" OR TS=”Velocardiofacial Syndrome” OR TS=”Sedlackova Syndrome” OR TS=”Shprintzen Syndrome” OR TS=rasopathies OR TS=”Langerhans cell histiocytosis” OR TS=”Immunodeficiency syndrome”) OR(TS=radiation OR TS=”ionizing radiation” OR TS=”radiation accident” OR TS=”nuclear accident” OR TS=”nuclear reactor accident” OR TS=”background radiation” OR TS=”natural radiation” OR TS=”medical radiation” OR TS=”atomic bomb” OR TS=”x-ray computed tomography” OR TS=”ct scan” OR TS=”ct imaging” OR TS=”computed tomography” OR TS=”pet-ct scanner” OR TS=”pet-ct scan” OR TS=”nuclear medicine” OR TS=”diagnostic radiation” OR TS=radiotherapy OR TS=”radiation treatment” OR TS=”pet scanner” OR TS=”pet scan” OR TS=”medical screening” OR TS=”occupational radiation”) OR (TS=leukaemia OR TS=lymphoma OR TS=”haematologic malignancy” OR TS=”haematologic disease” OR TS=”haematologic neoplasm” OR TS=”brain tumour” OR TS=”breast cancer” OR TS=”thyroid tumour” OR TS=”thyroid cancer” OR TS=”thyroid carcinoma” OR TS=”thyroid adenoma”) AND LANGUAGE: (English)			
(TS = (rb1 AND mutation) OR TS = (rb1 AND gene) OR TS = (retinoblastoma AND gene)) AND (TS = radiotherapy) AND (TS = retinoblastoma) AND LANGUAGE: (English) Indexes=SCI-EXPANDED, SSCI, A&HCI, CPCI-S, CPCI-SSH, ESCI Timespan=All years	Web of Science	133	2020-05-02
('disease predisposition':ti,ab,kw OR predisposing:ti,ab,kw OR radiosensitivity:ti,ab,kw OR 'genetic disorder':ti,ab,kw OR 'genetic disease':ti,ab,kw OR 'genetic susceptibility':ti,ab,kw OR 'tumor suppressor gene':ti,ab,kw OR 'brca1 protein':ti,ab,kw OR 'brca2 protein':ti,ab,kw OR	EMBASE	368 3176	2019-11-14 2020-02-12

'predisposing cancer syndrome':ti,ab,kw OR 'hereditary tumor syndrome':ti,ab,kw OR 'hereditary neoplastic syndrome':ti,ab,kw OR 'congenital malformation':ti,ab,kw OR 'congenital disorder':ti,ab,kw OR 'down syndrome':ti,ab,kw OR '47, xy, +21':ti,ab,kw OR 'trisomy g':ti,ab,kw OR '47, xx, +21':ti,ab,kw OR 'trisomy 21':ti,ab,kw OR 'noonan syndrome':ti,ab,kw OR 'turner syndrome':ti,ab,kw OR 'wiskott aldrich syndrome':ti,ab,kw OR imd2:ti,ab,kw OR 'Other phacomatoses':ti,ab,kw OR 'granular lymphocytes':ti,ab,kw OR 'immunodeficiency 2':ti,ab,kw OR 'ataxia telangiectasia':ti,ab,kw OR 'atm gene':ti,ab,kw OR 'louis bar syndrome':ti,ab,kw OR 'shwachman syndrome':ti,ab,kw OR 'bloom syndrome':ti,ab,kw OR 'monosomy 7':ti,ab,kw) AND 'bone marrow':ti,ab,kw 'myelodysplastic syndrome':ti,ab,kw AND 'monosomy 7':ti,ab,kw OR 'chromosome 7':ti,ab,kw OR 'myelodysplastic syndrome':ti,ab,kw OR 'chromosome disorder':ti,ab,kw OR 'fanconi anemia':ti,ab,kw OR 'fanconi hypoplastic anemia':ti,ab,kw OR 'bone marrow transplantation':ti,ab,kw OR 'hematopoietic stem cell':ti,ab,kw OR 'humoral immune deficiency':ti,ab,kw OR 'li fraumeni* syndrome':ti,ab,kw OR 'nijmegen breakage syndrome':ti,ab,kw OR 'nbs1 gene':ti,ab,kw OR neurofibromatosis:ti,ab,kw OR 'neurofibromatosis type 1':ti,ab,kw OR 'nf1 gene':ti,ab,kw OR 'molluscum fibrosum':ti,ab,kw OR 'neurofibromatosis type 2':ti,ab,kw OR 'nf2 gene':ti,ab,kw OR 'tuberous sclerosis':ti,ab,kw OR 'pelizaeus merzbacher disease':ti,ab,kw OR 'cerebral sclerosis':ti,ab,kw 'klinefelter syndrome':ti,ab,kw OR '47,xxy syndrome':ti,ab,kw OR '48, xxyy syndrome':ti,ab,kw OR '49,xxxxy syndrome':ti,ab,kw OR 'xxxy male':ti,ab,kw OR 'basal cell nevus syndrome':ti,ab,kw OR 'gorlin syndrome':ti,ab,kw OR 'von hippel lindau disease':ti,ab,kw OR 'vhl syndrome':ti,ab,kw OR 'turcot syndrome':ti,ab,kw OR 'Kostmann granulocytopenia':ti,ab,kw OR 'Nevoid basal cell syndrome':ti,ab,kw OR 'mmr deficiency':ti,ab,kw OR 'mismatch repair deficiency':ti,ab,kw OR 'mismatch repair cancer syndrome':ti,ab,kw OR 'human immunodeficiency virus':ti,ab,kw OR 'aids virus':ti,ab,kw OR 'acquired immune deficiency syndrome virus':ti,ab,kw OR 'severe combined immune deficiency':ti,ab,kw OR 'organ transplantation':ti,ab,kw OR 'organ donor':ti,ab,kw OR 'tissue transplantation':ti,ab,kw OR 'graft recipient':ti,ab,kw OR 'common variable immunodeficiency':ti,ab,kw OR 'congenital cardiovascular anomaly':ti,ab,kw OR 'congenital heart defects':ti,ab,kw OR 'nervous system malformation':ti,ab,kw OR 'central nervous system malformation':ti,ab,kw OR 'nervous system abnormality':ti,ab,kw OR 'congenital nervous system malformation':ti,ab,kw OR 'congenital malformation':ti,ab,kw OR 'birth defect':ti,ab,kw OR 'congenital abnormality':ti,ab,kw OR 'congenital anamoly':ti,ab,kw OR 'congenital disorder':ti,ab,kw OR '22q11 2 deletion syndrome':ti,ab,kw OR 'digeorge syndrome':ti,ab,kw OR (third:ti,ab,kw AND 'fourth pharyngeal pouch syndrome':ti,ab,kw) OR 'velocardiofacial syndrome':ti,ab,kw OR 'sedlackova syndrome':ti,ab,kw OR 'shprintzen syndrome':ti,ab,kw OR rasopathies:ti,ab,kw OR 'langerhans cell histiocytosis':ti,ab,kw OR 'immune deficiency':ti,ab,kw OR 'immunodeficiency syndrome':ti,ab,kw radiation:ti,ab,kw OR 'ionizing radiation':ti,ab,kw OR 'radiation accident':ti,ab,kw OR 'nuclear accident':ti,ab,kw OR 'nuclear reactor accident':ti,ab,kw OR 'background radiation':ti,ab,kw OR 'natural radiation':ti,ab,kw OR 'medical radiation':ti,ab,kw OR 'atomic bomb':ti,ab,kw OR radiography:ti,ab,kw OR 'x-ray computed tomography':ti,ab,kw OR 'computed tomography scanner':ti,ab,kw OR 'ct scan':ti,ab,kw OR 'ct imaging':ti,ab,kw OR 'computed tomography':ti,ab,kw OR 'pet-ct scanner':ti,ab,kw OR 'nuclear medicine':ti,ab,kw OR 'diagnostic radiation':ti,ab,kw OR radiotherapy:ti,ab,kw OR 'radiation treatment':ti,ab,kw OR 'pet scanner':ti,ab,kw OR 'medical screening':ti,ab,kw OR 'occupational radiation':ti,ab,kw leukemia:ti,ab,kw OR lymphoma:ti,ab,kw OR 'hematologic malignancy':ti,ab,kw OR 'hematologic disease':ti,ab,kw OR 'haematologic neoplasm':ti,ab,kw OR 'brain tumor':ti,ab,kw OR 'breast cancer':ti,ab,kw OR 'thyroid tumor':ti,ab,kw OR 'thyroid cancer':ti,ab,kw OR 'thyroid carcinoma':ti,ab,kw OR 'thyroid adenoma':ti,ab,kw			

('rb1 mutation':ti,ab,kw OR 'rb1 gene':ti,ab,kw OR 'retinoblastoma gene':ti,ab,kw) AND radiotherapy:ti,ab,kw AND retinoblastoma:ti,ab,kw	EMBASE	55	2020-05-02

Table S3. Summary of the risk of bias assessment for the included studies.

Study	Andrieu et al. 2006	Berrington de Gonzalez et al. 2016	Damiola et al. 2014	Esther et al. 2007	Esther et al. 2013	Pijpe et al. 2012	Journy et al. 2015 Journy et al. 2016
Recruitment							
YES (low risk of bias)	Recruitment from various countries independent of radiation exposure	Study participants were recruited from the same population and the same time frame	Study participants were recruited from the same population and the same time frame	Study participants were recruited from the same populations and the same time frame	Study participants were recruited from the same populations and the same time frame	Study participants were recruited from the same population and the same time frame	Study participants were recruited from the same population and the same time frame
Blinding (Performance bias)							
YES (low risk of bias)	Self-administered questionnaire - blinding not possible	Blinding not applicable in a record linkage study. Exposure was unknown in selection of Cohort and Dosimetrists were blinded to case status	Blinding not applicable in this case-control. Dosimetrists were blinded to case/control status	Self-administered questionnaire - blinding not possible	Self-administered questionnaire - blinding not possible	Self-administered questionnaire - blinding not possible	Blinding not applicable in a record linkage study. Exposure was unknown in selection of Cohort and Dosimetrists were blinded to case status
Exposure assessment (performance bias)							
YES (low risk of bias)		Dose reconstruction based on data extracted from Radiological Information System. Low risk of substantial misclassification	Sound dose-reconstruction method (Drozdovitch et al) - with little risk of substantial exposure misclassification				Dose reconstruction based on data extracted from Radiological Information System. Low risk of substantial misclassification
Probably YES	Chest x-ray exposure determined through questionnaire and not validated with medical records. Potential for recall bias though it is likely to be small in this population of women who are already at high risk for BC due to their mutation			Chest x-ray exposure determined through questionnaire and not validated with medical records. Potential for recall bias though it is likely to be small in this population of women who are already at high risk for BC due to their mutation.	Chest x-ray exposure determined through questionnaire and not validated with medical records. Potential for recall bias though it is likely to be small in this population of women who are already at high risk for BC due to their mutation.	Diagnostic IR exposure determined through questionnaire and not validated with medical records. Potential for recall bias though likely to be small in this population of women who are at high risk for BC due to their mutation.	
Confounding							
YES (low risk of bias)	Parity, oophorectomy, family membership, year of birth and country considered. Sub-cohort analysis to consider possible survival bias and adjustment for potential surveillance bias		Adjusted for age and sex, iodine deficiency and supplementation. Information available on potential thyroid cancer risk factors		Adjusted for age, country/region, family history of breast or ovarian cancer and number of full-term pregnancies. Analyses restricted to sub cohort to account for possible survival bias and adjusted for potential surveillance bias	Adjusted for age at entry in sub cohort, parity and menopause (including surgical). Adjustment for potential surveillance bias	
Probably YES		Possible confounding by reverse causation, though doses lagged to reduce this and search through medical records for possible missing cancer diagnoses. Adjusted for attained age, age at exposure, gender and SES. Subjects with previous cancers were censored at first cancer		Parity not considered. Adjusted for age, study center, country of birth, education and for possible surveillance bias			Possible confounding by reverse causation, though doses lagged to reduce this. Adjusted for attained age, age at exposure and gender

Study	Andrieu et al. 2006	Berrington de Gonzalez et al. 2016	Damiola et al. 2014	Esther et al. 2007	Esther et al. 2013	Pijpe et al. 2012	Journy et al. 2015 Journy et al. 2016
Outcome assessment (Detection bias)							
YES (low risk of bias)	Cases ascertained through questionnaire and linkage to population-based cancer registries (Goldgar, D., Bonnardel, C., Renard, H. et al. The International BRCA1 /2Carrier Cohort Study: Purpose, Rationale, and Study Design.Breast Cancer Res 2, E010 (2000) https://doi.org/10.1186/bcr93)	Cases ascertained through linkage with UK Cancer registry, estimated to be 90% complete	Cases ascertained retrospectively from cancer registries and prospectively from records of regional oncology dispensaries, surgical departments of central hospitals and, in Belarus, the Republican Scientific and Practical Center for Thyroid Tumors in Minsk, where most young people diagnosed with thyroid cancer in Belarus (Cardis et al 2005)	Cases ascertained through population-based cancer registries in Australia and Canada. Completeness is high	Cases ascertained through population-based cancer registries in Australia and Canada. Completeness is high	Cases ascertained through linkage with population-based cancer registries and medical record. Completeness expected to be high	Cases ascertained through linkage with the French National Cancer registry (RNCE) - from 1990 for haematological malignancies and 2000 for other malignancies in children 0-15. Completeness good
Selective reporting							
YES (low risk of bias)	All outcomes outlined have been reported	All outcomes outlined have been reported	All outcomes outlined have been reported	All outcomes outlined have been reported	All outcomes outlined have been reported	All outcomes outlined have been reported	All outcomes outlined have been reported
Other bias							
YES (low risk of bias)		No other source of bias	No other source of bias				No other source of bias
Probably YES	Possible surveillance bias - analyses corrected for this (see Confounding)			Possible surveillance bias - analyses corrected for this (see Confounding)	Possible surveillance bias - analyses corrected for this (see Confounding)	Possible surveillance bias - analyses corrected for this (see Confounding)	
Conflict of interest							
YES (low risk of bias)	No apparent conflict of interest demonstrated or suspected	No apparent conflict of interest demonstrated or suspected	No apparent conflict of interest demonstrated or suspected	No apparent conflict of interest demonstrated or suspected	No apparent conflict of interest demonstrated or suspected	No apparent conflict of interest demonstrated or suspected	No apparent conflict of interest demonstrated or suspected

Table S3 (continued). Summary of the risk of bias assessment for the included studies.

Study	Kleinerman et al. 2005	Lecarpentier et al. 2011	Little et al. 2014	Meulepas et al. 2019	Momani et al. 2004	Nikkilä et al. 2018	Gronwald et al. 2008
Recruitment							
YES (low risk of bias)	Study participants were recruited from the same populations and the same time frame	Study participants were recruited from the same population and the same time frame	Study participants were recruited from the same populations and the same time frame	Study participants were recruited from the same population and the same time frame	Study participants were recruited from the same population and the same time frame	Study participants were recruited from the same population and the same time frame. PF, obtained through outpatient register data	Study participants were recruited from the same population and the same time frame
Blinding (Performance bias)							
YES (low risk of bias)	Blinding not applicable in a record linkage study. Exposure was unknown in selection of Cohort and Dosimetrists were blinded to case status	Self-administered questionnaire - blinding not possible	Blinding not applicable in a record linkage study. Exposure was unknown in selection of Cohort and Dosimetrists were blinded to case status	Blinding not applicable in a record linkage study. Exposure was unknown in selection of Cohort and Dosimetrists were blinded to case status	Blinding not applicable in a record linkage study. Exposure was unknown in selection of Cohort and Dosimetrists were blinded to case status	Blinding not applicable in a record linkage study. Exposure was unknown in selection of Cohort and Dosimetrists were blinded to case status	Blinding not applicable in a record linkage study. Exposure was unknown in selection of Cohort and Dosimetrists were blinded to case status
Exposure assessment (performance bias)							
YES (low risk of bias)	No individual dose reconstruction - exposed vs non-exposed. No reason to expect substantial misclassification		Dose estimation based on radiotherapy treatment details from records and detailed modelling. No reason to suspect substantial misclassification	Dose reconstruction based on data extracted from Radiological Information System. Low risk of substantial misclassification	Dose estimation based on radiotherapy treatment details from records and modelling (Schneider et al 1993). Some misclassification expected but no reason to suspect substantial misclassification	Dose reconstruction based on data extracted from Radiological Information System. Low risk of substantial misclassification	
Probably YES		Chest x-ray exposure determined through questionnaire and not validated with medical records. Potential for recall bias though it is likely to be small in this population of women who are already at high risk for BC due to their mutation.					Chest x-ray exposure determined through questionnaire and not validated with medical records. Potential for recall bias though it is likely to be small in this population of women who are already at high risk for BC due to their mutation.
Confounding							
YES (low risk of bias)		Analyses adjusted for menopausal status, parity, BMI, gene, alcohol and tobacco consumption and for possible surveillance bias					
Probably YES	Parity and education not considered. Analyses adjusted for age, sex and calendar year		Parity not considered. Adjusted for age, sex, birth cohort, blindness and chemotherapy	Possible confounding by reverse causation, though doses lagged to reduce this. Adjusted for attained age, age at exposure, SES and gender	Possible surveillance bias after thyroid cancer in first sibling	Possible confounding by reverse causation, though doses lagged to reduce this. Age, sex, gestational age, maternal smoking during pregnancy, background radiation, parental education and parental socioeconomic status were considered	No adjustment for important risk factors for BC. Age adjustment unclear
Outcome assessment (Detection bias)							
YES (low risk of bias)		Cases ascertained from family cancer clinics		Cases ascertained through linkage with national cancer registry		Cancer cases identified through Finnish Cancer Registry	Cases ascertained from BC treatment centers throughout Poland

Study	Kleinerman et al. 2005	Lecarpentier et al. 2011	Little et al. 2014	Meulepas et al. 2019	Momani et al. 2004	Nikkilä et al. 2018	Gronwald et al. 2008
Probably YES	New cases were identified through contact with the study subjects or their families. Diagnoses were confirmed via pathology reports, hospital/physician records, death certificates and autopsy reports. Completeness of case ascertainment not mentioned		See Kleinerman et al 2005 for outcome ascertainment for US Cohort. Outcome ascertainment for UK Cohort ok: national population-based cancer registry		New cases identified from follow-up questionnaires. Diagnoses validated but completeness not mentioned		
Selective reporting							
YES (low risk of bias)	All outcomes outlined have been reported	All outcomes outlined have been reported	All outcomes outlined have been reported	All outcomes outlined have been reported	All outcomes outlined have been reported	All outcomes outlined have been reported	All outcomes outlined have been reported
Other bias							
YES (low risk of bias)	No other source of bias		No other source of bias	No other source of bias	No other source of bias	No other source of bias	No other source of bias
Probably YES		Possible surveillance bias - analyses corrected for this (see Confounding)					
Conflict of interest							
YES (low risk of bias)	No apparent conflict of interest demonstrated or suspected	No apparent conflict of interest demonstrated or suspected	No apparent conflict of interest demonstrated or suspected	No apparent conflict of interest demonstrated or suspected	No apparent conflict of interest demonstrated or suspected	No apparent conflict of interest demonstrated or suspected	No apparent conflict of interest demonstrated or suspected

Table S4. Reason for exclusion during full-text evaluation (*n*=94)

Reference	Reason for exclusion
Caspi, O., Polliack, A., Klar, R., & Ben-Yehuda, D. (1995). The association of inflammatory bowel disease and leukemia--coincidence or not?. Leukemia & lymphoma, 17(3-4), 255–262. https://doi.org/10.3109/10428199509056830	Population
Alapetite, C., Cosset, J. M., Bourguignon, M. H., & Masse, R. (2000). Genetic susceptibility to radiations. Which impact on medical practice? The quarterly journal of nuclear medicine: official publication of the Italian Association of Nuclear Medicine (AIMN) [and] the International Association of Radiopharmacology (IAR), 44(4), 347–354.	Study design
Broeks, A., Russell, N. S., Floore, A. N., Urbanus, J. H., Dahler, E. C., van T Veer, M. B., Hagenbeek, A., Noordijk, E. M., Crommelin, M. A., van Leeuwen, F. E., & van T Veer, L. J. (2000). Increased risk of breast cancer following irradiation for Hodgkin's disease is not a result of ATM germline mutations. International journal of radiation biology, 76(5), 693–698. https://doi.org/10.1080/095530000138367	Population/Outcome
Kozanoğlu, B. , Sezgin, G. , Bagır, E. , Özkan, A. , Bayram, İ. , Tanyeli, A. , Küpeli, S. (2015), Secondary Lymphoma in a Patient with a History of Unilateral Retinoblastoma. Cukurova Medical Journal 40, 33-37 https://doi.org/10.17826/cutf.94994	Study design
Brantley, M. A., Jr, Worley, L., & Harbour, J. W. (2002). Altered expression of Rb and p53 in uveal melanomas following plaque radiotherapy. American journal of ophthalmology, 133(2), 242–248. https://doi.org/10.1016/s0002-9394(01)01362-9	Population
Mitsuhashi, M., Peel, D., Ziogas, A., & Anton-Culver, H. (2009). Enhanced Expression of Radiation-induced Leukocyte CDKN1A mRNA in Multiple Primary Breast Cancer Patients: Potential New Marker of Cancer Susceptibility. Biomarker insights, 4, 201–209. https://doi.org/10.4137/bmi.s3774	Population
Eng, C., Li, F. P., Abramson, D. H., Ellsworth, R. M., Wong, F. L., Goldman, M. B., Seddon, J., Tarbell, N., & Boice, J. D., Jr (1993). Mortality from second tumors among long-term survivors of retinoblastoma. Journal of the National Cancer Institute, 85(14), 1121–1128. https://doi.org/10.1093/jnci/85.14.1121	Population/Outcome
Tucci, F., & Aricò, M. (2008). Treatment of pediatric acute lymphoblastic leukemia. Haematologica, 93(8), 1124–1128. https://doi.org/10.3324/haematol.13517	Exposure
Gault, N., Verbiest, T., Badie, C., Romeo, P. H., & Bouffler, S. (2019). Hematopoietic stem and progenitor cell responses to low radiation doses - implications for leukemia risk. International journal of radiation biology, 95(7), 892–899. https://doi.org/10.1080/09553002.2019.1569777	Population
Hall, J., & Angèle, S. (1999). Radiation, DNA damage and cancer. Molecular medicine today, 5(4), 157–164. https://doi.org/10.1016/s1357-4310(99)01435-5	Study design
Spunt, S. L., Rodriguez-Galindo, C., Fuller, C. E., Harper, J., Krasin, M. J., Billups, C. A., & Khoury, J. D. (2006). Ewing sarcoma-family tumors that arise after treatment of primary childhood cancer. Cancer, 107(1), 201–206. https://doi.org/10.1002/cncr.21962	Population
Nikkilä, A., Erme, S., Arvela, H., Holmgren, O., Raitanen, J., Lohi, O., & Auvinen, A. (2016). Background radiation and childhood leukemia: A nationwide register-based case-control study. International journal of cancer, 139(9), 1975–1982. https://doi.org/10.1002/ijc.30264	Population

Sankaranarayanan, K., & Chakraborty, R. (1995). Cancer predisposition, radiosensitivity and the risk of radiation-induced cancers. I. Background. Radiation research, 143(2), 121–143.	Study design
Goldfrank, D., Chuai, S., Bernstein, J. L., Ramon Y Cajal, T., Lee, J. B., Alonso, M. C., Diez, O., Baiget, M., Kauff, N. D., Offit, K., & Robson, M. (2006). Effect of mammography on breast cancer risk in women with mutations in BRCA1 or BRCA2. Cancer epidemiology, biomarkers & prevention : a publication of the American Association for Cancer Research, cosponsored by the American Society of Preventive Oncology, 15(11), 2311–2313. https://doi.org/10.1158/1055-9965.EPI-06-0176	Population
Ma, Y. P., van Leeuwen, F. E., Cooke, R., Broeks, A., Enciso-Mora, V., Olver, B., Lloyd, A., Broderick, P., Russell, N. S., Janus, C., Ashworth, A., Houlston, R. S., & Swerdlow, A. J. (2012). FGFR2 genotype and risk of radiation-associated breast cancer in Hodgkin lymphoma. Blood, 119(4), 1029–1031. https://doi.org/10.1182/blood-2011-10-383380	Exposure
Curwen, G. B., Winther, J. F., Tawn, E. J., Smart, V., Whitehouse, C. A., Rees, G. S., Olsen, J. H., Guldberg, P., Rechnitzer, C., Schrøder, H., Bryant, P. E., Sheng, X., Lee, H. S., Chakraborty, R., & Boice, J. D. (2005). G(2) chromosomal radiosensitivity in Danish survivors of childhood and adolescent cancer and their offspring. British journal of cancer, 93(9), 1038–1045. https://doi.org/10.1038/sj.bjc.6602807	Population/Exposure
Temming, P., Arendt, M., Viehmann, A., Eisele, L., Le Guin, C. H., Schündeln, M. M., Biewald, E., Astrahantseff, K., Wieland, R., Bornfeld, N., Sauerwein, W., Eggert, A., Jöckel, K. H., & Lohmann, D. R. (2017). Incidence of second cancers after radiotherapy and systemic chemotherapy in heritable retinoblastoma survivors: A report from the German reference center. Pediatric blood & cancer, 64(1), 71–80. https://doi.org/10.1002/pbc.26193	Population/Outcome
Sigurdson, A. J., & Stram, D. O. (2012). Genetic predisposition to radiation-related cancer and potential implications for risk assessment. Annals of the ICRP, 41(3-4), 108–116. https://doi.org/10.1016/j.icrp.2012.06.030	Population
Abstracts: AACR Special Conference: Cancer Susceptibility and Cancer Susceptibility Syndromes (2014)	Missing
Janin, N., Andrieu, N., Ossian, K., Laugé, A., Croquette, M. F., Griscelli, C., Debré, M., Bressac-de-Paillerets, B., Aurias, A., & Stoppa-Lyonnet, D. (1999). Breast cancer risk in ataxia telangiectasia (AT) heterozygotes: haplotype study in French AT families. British journal of cancer, 80(7), 1042–1045. https://doi.org/10.1038/sj.bjc.6690460	Exposure
Olsen, M., Garne, E., Sværke, C., Søndergaard, L., Nissen, H., Andersen, H. Ø., Hjortdal, V. E., Johnsen, S. P., & Videbæk, J. (2014). Cancer risk among patients with congenital heart defects: a nationwide follow-up study. Cardiology in the young, 24(1), 40–46. https://doi.org/10.1017/S1047951112002144	Exposure
Allan J. M. (2008). Genetic susceptibility to radiogenic cancer in humans. Health physics, 95(5), 677–686. https://doi.org/10.1097/01.HP.0000326339.06405.ea	Study design
Bourguignon, M. H., Gisone, P. A., Perez, M. R., Michelin, S., Dubner, D., Giorgio, M. D., & Carosella, E. D. (2005). Genetic and epigenetic features in radiation sensitivity Part I: cell signalling in radiation response. European journal of nuclear medicine and molecular imaging, 32(2), 229–246. https://doi.org/10.1007/s00259-004-1730-7	Study design
Vijapura, C., Saad Aldin, E., Capizzano, A. A., Policeni, B., Sato, Y., & Moritani, T. (2017). Genetic Syndromes Associated with Central Nervous System Tumors. Radiographics : a review publication of the Radiological	Study design

Society of North America, Inc, 37(1), 258–280. https://doi.org/10.1148/rg.2017160057	
Berrington de Gonzalez, A., Berg, C. D., Visvanathan, K., & Robson, M. (2009). Estimated risk of radiation-induced breast cancer from mammographic screening for young BRCA mutation carriers. Journal of the National Cancer Institute, 101(3), 205–209. https://doi.org/10.1093/jnci/djn440	Study design
Hauptmann, M., & Meulepas, J. M. (2012). CT scans in childhood and risk of leukaemia and brain tumours. Lancet (London, England), 380(9855), 1736–1737. https://doi.org/10.1016/S0140-6736(12)61983-7	No primary data
Jongmans, W., & Hall, J. (1999). Cellular responses to radiation and risk of breast cancer. European journal of cancer (Oxford, England : 1990), 35(4), 540–548. https://doi.org/10.1016/s0959-8049(98)00414-6	Study design
Apsalikov, B., Manambaeva, Z., Ospanov, E., Massabayeva, M., Zhabagin, K., Zhagiparova, Z., Maximov, V., Voropaeva, E., Apsalikov, K., Belikhina, T., Abdrahmanov, R., Cherepkova, E., Tanatarov, S., Massadykov, A., & Urazalina, N. (2016). BRCA1 and TP53 Gene-Mutations: Family Predisposition and Radioecological Risk of Developing Breast Cancer. Asian Pacific journal of cancer prevention: APJCP, 17(8), 4059–4062.	Population
Reiner, A. S., Sisti, J., John, E. M., Lynch, C. F., Brooks, J. D., Mellemkjær, L., Boice, J. D., Knight, J. A., Concannon, P., Capanu, M., Tischkowitz, M., Robson, M., Liang, X., Woods, M., Conti, D. V., Duggan, D., Shore, R., Stram, D. O., Thomas, D. C., Malone, K. E., … Bernstein, J. L. (2018). Breast Cancer Family History and Contralateral Breast Cancer Risk in Young Women: An Update From the Women's Environmental Cancer and Radiation Epidemiology Study. Journal of clinical oncology: official journal of the American Society of Clinical Oncology, 36(15), 1513–1520. https://doi.org/10.1200/JCO.2017.77.3424	Population
Tucker M. (2000). Major cancer susceptibility genes and radiation: what do we know? Radiation research, 154(6), 740–741.	Study design
de Vathaire, F., Hardiman, C., Shamsaldin, A., Campbell, S., Grimaud, E., Hawkins, M., Raquin, M., Oberlin, O., Diallo, I., Zucker, J. M., Panis, X., Lagrange, J. L., Daly-Schveitzer, N., Lemerle, J., Chavaudra, J., Schlumberger, M., & Bonaïti, C. (1999). Thyroid carcinomas after irradiation for a first cancer during childhood. Archives of internal medicine, 159(22), 2713–2719. https://doi.org/10.1001/archinte.159.22.2713	Population
Allan J. M. (2019). Genetic susceptibility to breast cancer in lymphoma survivors. Blood, 133(10), 1004–1006. https://doi.org/10.1182/blood-2019-01-894279	Population/ No primary data
Hendrickson, P. G., Luo, Y., Kohlmann, W., Schiffman, J., Maese, L., Bishop, A. J., Lloyd, S., Kokeny, K. E., Hitchcock, Y. J., Poppe, M. M., Gaffney, D. K., & Tao, R. (2020). Radiation therapy and secondary malignancy in Li-Fraumeni syndrome: A hereditary cancer registry study. Cancer medicine, 9(21), 7954–7963. https://doi.org/10.1002/cam4.3427	Population
Feng, Y., Spezia, M., Huang, S., Yuan, C., Zeng, Z., Zhang, L., Ji, X., Liu, W., Huang, B., Luo, W., Liu, B., Lei, Y., Du, S., Vuppalapati, A., Luu, H. H., Haydon, R. C., He, T. C., & Ren, G. (2018). Breast cancer development and progression: Risk factors, cancer stem cells, signaling pathways, genomics, and molecular pathogenesis. Genes & diseases, 5(2), 77–106. https://doi.org/10.1016/j.gendis.2018.05.001	Study design
Genetic susceptibility to cancer. ICRP publication 79. Approved by the Commission in May 1997. International Commission on Radiological Protection. (1998). Annals of the ICRP, 28(1-2), 1–157.	Study design

Leong, T., Whitty, J., Keilar, M., Mifsud, S., Ramsay, J., Birrell, G., Venter, D., Southey, M., & McKay, M. (2000). Mutation analysis of BRCA1 and BRCA2 cancer predisposition genes in radiation hypersensitive cancer patients. International journal of radiation oncology, biology, physics, 48(4), 959–965. https://doi.org/10.1016/s0360-3016(00)00728-8	Population/Outcome
Yard, B. D., Adams, D. J., Chie, E. K., Tamayo, P., Battaglia, J. S., Gopal, P., Rogacki, K., Pearson, B. E., Phillips, J., Raymond, D. P., Pennell, N. A., Almeida, F., Cheah, J. H., Clemons, P. A., Shamji, A., Peacock, C. D., Schreiber, S. L., Hammerman, P. S., & Abazeed, M. E. (2016). A genetic basis for the variation in the vulnerability of cancer to DNA damage. Nature communications, 7, 11428. https://doi.org/10.1038/ncomms11428	Population
Boice J. D., Jr (2015). Radiation epidemiology and recent paediatric computed tomography studies. Annals of the ICRP, 44(1 Suppl), 236–248. https://doi.org/10.1177/0146645315575877	Study design
Broeks, A., Braaf, L. M., Huseinovic, A., Nooijen, A., Urbanus, J., Hogervorst, F. B., Schmidt, M. K., Klijn, J. G., Russell, N. S., Van Leeuwen, F. E., & Van 't Veer, L. J. (2007). Identification of women with an increased risk of developing radiation-induced breast cancer: a case only study. Breast cancer research: BCR, 9(2), R26. https://doi.org/10.1186/bcr1668	Population
Westwood, T. D., Hogan, C., Julyan, P. J., Coutts, G., Bonington, S., Carrington, B., Taylor, B., Khoo, S., & Bonington, A. (2013). Utility of FDG-PETCT and magnetic resonance spectroscopy in differentiating between cerebral lymphoma and non-malignant CNS lesions in HIV-infected patients. European journal of radiology, 82(8), e374–e379. https://doi.org/10.1016/j.ejrad.2013.03.008	Population/Outcome
Marees, T., Moll, A. C., Imhof, S. M., de Boer, M. R., Ringens, P. J., & van Leeuwen, F. E. (2008). Risk of second malignancies in survivors of retinoblastoma: more than 40 years of follow-up. Journal of the National Cancer Institute, 100(24), 1771–1779. https://doi.org/10.1093/jnci/djn394	Exposure
Chaussade, A., Millot, G., Wells, C., Brisse, H., Laé, M., Savignoni, A., Desjardins, L., Dendale, R., Doz, F., Aerts, I., Jimenez, I., Cassoux, N., Stoppa Lyonnet, D., Gauthier Villars, M., & Houdayer, C. (2019). Correlation between RB1germline mutations and second primary malignancies in hereditary retinoblastoma patients treated with external beam radiotherapy. European journal of medical genetics, 62(3), 217–223. https://doi.org/10.1016/j.ejmg.2018.07.017	Exposure/Outcome
Genetic susceptibility to cancer. ICRP publication 79. Approved by the Commission in May 1997. International Commission on Radiological Protection. (1998). Annals of the ICRP, 28(1-2), 1–157.	Duplicate
Temming, P., Viehmann, A., Arendt, M., Eisele, L., Spix, C., Bornfeld, N., Sauerwein, W., Jöckel, K. H., & Lohmann, D. R. (2015). Pediatric second primary malignancies after retinoblastoma treatment. Pediatric blood & cancer, 62(10), 1799–1804. https://doi.org/10.1002/pbc.25576	Exposure/Outcome
Kony, S. J., de Vathaire, F., Chompret, A., Shamsaldim, A., Grimaud, E., Raquin, M. A., Oberlin, O., Brugières, L., Feunteun, J., Eschwège, F., Chavaudra, J., Lemerle, J., & Bonaïti-Pellié, C. (1997). Radiation and genetic factors in the risk of second malignant neoplasms after a first cancer in childhood. Lancet (London, England), 350(9071), 91–95. https://doi.org/10.1016/S0140-6736(97)01116-1	Population/Outcome
Hill, J. W., Tansavatdi, K., Lockett, K. L., Allen, G. O., Takita, C., Pollack, A., & Hu, J. J. (2009). Validation of the cell cycle G(2) delay assay in assessing ionizing radiation sensitivity and breast cancer risk. Cancer management and research, 1, 39–48. https://doi.org/10.2147/cmar.s4548	Population
Balaguer J.; Harto M.; Serra I.; Oltra S.; Ãndez M.H.; Castel V. (2009) Mortality from second tumors among long-term survivors of retinoblastoma: An 87-case series report. Pediatric Blood and Cancer. 53, 881	Study design

Richardson R. B. (2011). Stem cell niches and other factors that influence the sensitivity of bone marrow to radiation-induced bone cancer and leukaemia in children and adults. International journal of radiation biology, 87(4), 343–359. https://doi.org/10.3109/09553002.2010.537430	Population/Outcome
Ullrich N. J. (2008). Inherited disorders as a risk factor and predictor of neurodevelopmental outcome in pediatric cancer. Developmental disabilities research reviews, 14(3), 229–237. https://doi.org/10.1002/ddrr.30	Study design
Ladouceur, A., Bernatsky, S., Ramsey-Goldman, R., & Clarke, A. E. (2018). Managing cancer risk in patients with systemic lupus erythematous. Expert review of clinical immunology, 14(10), 793–802. https://doi.org/10.1080/1744666X.2018.1519394	Exposure
Opstal-van Winden, A., de Haan, H. G., Hauptmann, M., Schmidt, M. K., Broeks, A., Russell, N. S., Janus, C., Krol, A., van der Baan, F. H., De Bruin, M. L., van Eggermond, A. M., Dennis, J., Anton-Culver, H., Haiman, C. A., Sawyer, E. J., Cox, A., Devilee, P., Hooning, M. J., Peto, J., Couch, F. J., … van Leeuwen, F. E. (2019). Genetic susceptibility to radiation-induced breast cancer after Hodgkin lymphoma. *Blood*, *133*(10), 1130–1139. https://doi.org/10.1182/blood-2018-07-862607	Population/Exposure
Bernstein, J. L., Thomas, D. C., Shore, R. E., Robson, M., Boice, J. D., Jr, Stovall, M., Andersson, M., Bernstein, L., Malone, K. E., Reiner, A. S., Lynch, C. F., Capanu, M., Smith, S. A., Tellhed, L., Teraoka, S. N., Begg, C. B., Olsen, J. H., Mellemkjaer, L., Liang, X., Diep, A. T., … WECARE Study Collaborative Group (2013). Contralateral breast cancer after radiotherapy among BRCA1 and BRCA2 mutation carriers: a WECARE study report. *European journal of cancer (Oxford, England : 1990)*, *49*(14), 2979–2985. https://doi.org/10.1016/j.ejca.2013.04.028	Population
Mathews, J. D., Forsythe, A. V., Brady, Z., Butler, M. W., Goergen, S. K., Byrnes, G. B., Giles, G. G., Wallace, A. B., Anderson, P. R., Guiver, T. A., McGale, P., Cain, T. M., Dowty, J. G., Bickerstaffe, A. C., & Darby, S. C. (2013). Cancer risk in 680,000 people exposed to computed tomography scans in childhood or adolescence: data linkage study of 11 million Australians. *BMJ (Clinical research ed.)*, *346*, f2360. https://doi.org/10.1136/bmj.f2360	Population
Snijders, A. M., Marchetti, F., Bhatnagar, S., Duru, N., Han, J., Hu, Z., Mao, J. H., Gray, J. W., & Wyrobek, A. J. (2012). Genetic differences in transcript responses to low-dose ionizing radiation identify tissue functions associated with breast cancer susceptibility. *PloS one*, *7*(10), e45394. https://doi.org/10.1371/journal.pone.0045394	Population
Zebisch, A., Lal, R., Müller, M., Lind, K., Kashofer, K., Girschikofsky, M., Fuchs, D., Wölfler, A., Geigl, J. B., & Sill, H. (2016). Acute myeloid leukemia with TP53 germ line mutations. *Blood*, *128*(18), 2270–2272. https://doi.org/10.1182/blood-2016-08-732610	Population/Exposure
Stevens, R. G., Morris, J. E., & Anderson, L. E. (2000). Hemochromatosis heterozygotes may constitute a radiation-sensitive subpopulation. *Radiation research*, *153*(6), 844–847.	No primary data
Someya, M., Sakata, K., Tauchi, H., Matsumoto, Y., Nakamura, A., Komatsu, K., & Hareyama, M. (2006). Association of ionizing radiation-induced foci of NBS1 with chromosomal instability and breast cancer susceptibility. *Radiation research*, *166*(4), 575–582. https://doi.org/10.1667/RR0638.1	Population
Hu, J. J., Smith, T. R., Miller, M. S., Lohman, K., & Case, L. D. (2002). Genetic regulation of ionizing radiation sensitivity and breast cancer risk. *Environmental and molecular mutagenesis*, *39*(2-3), 208–215. https://doi.org/10.1002/em.10058	Population
Akkineni Veena, Maneya Padma, Nuthan Kumar, Arun Kumar, L. Appaji, Aruna Kumari, (2019) Genetic syndromes in childhood cancers – A regional	Study design

cancer center experience, Pediatric Hematology Oncology Journal, Volume 4, Issue 2, Supplement, Page S15, https://doi.org/10.1016/j.phoj.2019.08.037.	
Chakraborty, R., Little, M. P., & Sankaranarayanan, K. (1997). Cancer predisposition, radiosensitivity and the risk of radiation-induced cancers. III. Effects of incomplete penetrance and dose-dependent radiosensitivity on cancer risks in populations. *Radiation research*, *147*(3), 309–320.	Population
Alsubhi, N., Middleton, F., Abdel-Fatah, T. M., Stephens, P., Doherty, R., Arora, A., Moseley, P. M., Chan, S. Y., Aleskandarany, M. A., Green, A. R., Rakha, E. A., Ellis, I. O., Martin, S. G., Curtin, N. J., & Madhusudan, S. (2016). Chk1 phosphorylated at serine345 is a predictor of early local recurrence and radio-resistance in breast cancer. *Molecular oncology*, *10*(2), 213–223. https://doi.org/10.1016/j.molonc.2015.09.009	Population
Briani, C., Schlotter, M., Lichter, P., & Kalla, C. (2006). Development of a mantle cell lymphoma in an ATM heterozygous woman after occupational exposure to ionising radiation and somatic mutation of the second allele. *Leukemia research*, *30*(9), 1193–1196. https://doi.org/10.1016/j.leukres.2005.11.010	Missing
Guillem, V. M., Collado, M., Terol, M. J., Calasanz, M. J., Esteve, J., Gonzalez, M., Sanzo, C., Nomdedeu, J., Bolufer, P., Lluch, A., & Tormo, M. (2007). Role of MTHFR (677, 1298) haplotype in the risk of developing secondary leukemia after treatment of breast cancer and hematological malignancies. *Leukemia*, *21*(7), 1413–1422. https://doi.org/10.1038/sj.leu.2404709	Population
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Allan, J. M., & Rabkin, C. S. (2005). Genetic susceptibility to iatrogenic malignancy. *Pharmacogenomics*, *6*(6), 615–628. https://doi.org/10.2217/14622416.6.6.615	Study design
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